

AI Computing in Large-Scale Era : Pre-trillion-scale Neural Network Models and Exa-scale Supercomputing

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Abstract—The development of AI computing has reached a critical inflection point. The scale of large-scale AI neural network model parameters has grown rapidly to "pre-trillion-scale" level. The computing needs of training large-scale AI neural network models have reached "exa-scale" level. Besides, AI Foundation Model also affects the correctness of AI applications, and becoming a new information security issue. Future AI development will be pushed by progress of computing power (supercomputer), algorithm (neural network model and parameter scale), and application (foundation model and downstream fine tuning). In particular, the computational efficiency of AI will be a key factor in the commercialization and popularization of AI applications.

Keywords—AI computing, Large-scale model, Large-scale Language model, Exa-scale Supercomputing, Foundation Model

I. PARAMETERS OF NEURAL NETWORK MODEL

The neural network architecture of AI has made great progress recently. Especially after the rise of the Transformer architecture [1], it triggered the growth of large scale neural network models. The number of parameters of the neural network has grown from 60 million in Alexnet [2] to 175 billion in OpenAI GPT-3 [3], and some researchers have begun to explore trillion-parameters neural network models.

The parameter of mainstream neural network architectures published in the past two years is between 65 Bn and 540 Bn, which is only one order smaller than the trillion-scale neural network, so it can be called pre-trillion-scale neural network model. See Fig.1 for details.

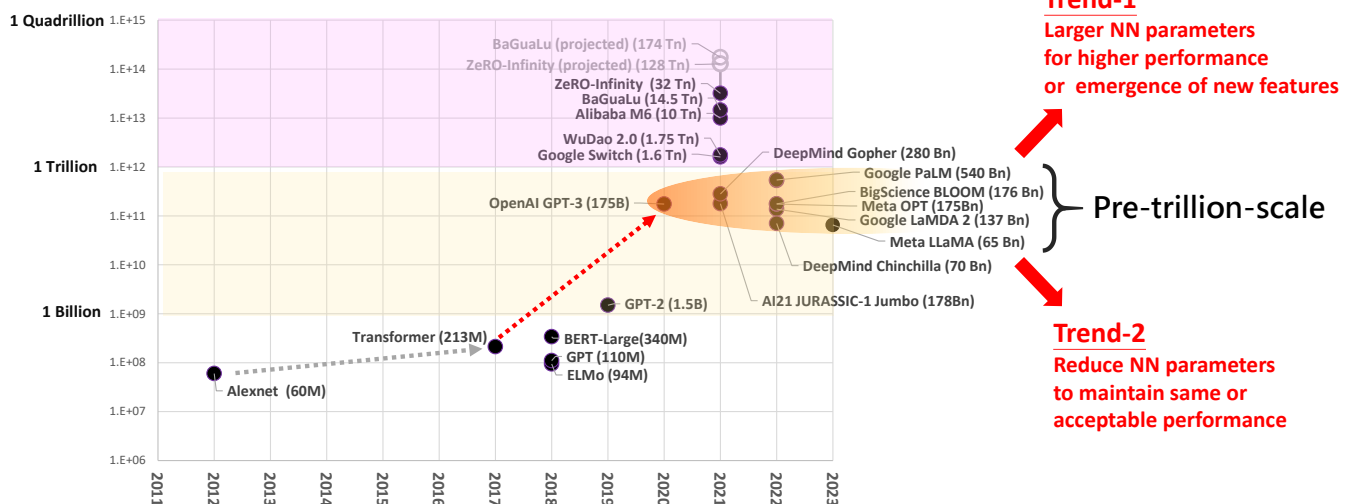
Because of the development of LLM (Large Language Model) and Generative AI, there are many amazing new AI applications, such as text to generate high-quality images, videos, music or even programming. Recently, ChatGPT, a conversational AI, has attracted hundreds of millions of users within two months. It demonstrated the potential of AI applications to impact on the way humans work and live.

II. AI COMPUTING REQUIREMENT

However, we need huge amount of computing power to training large neural networks in order to make the AI capabilities to emerge [4]. The number of parameters of the neural network has grown from 60M in Alexnet to 175 Bn in OpenAI GPT-3. Within seven years, the number of parameters has increased by 2917 times, and the computing requirements for training neural networks have increased by 300,000 times. This speed far exceeds the growth rate of Moore's Law of semiconductors, and pushes AI computing into the Large-Scale Era, as illustrated in Fig.2 [5].

Fig.1. Parameter Numbers of Neural Network (NN) Model over Time

Number of NN Parameters



III. PROGRESS OF SUPERCOMPUTING

There is a strong demand of exa-scale supercomputers to provide AI computing. However, it's not necessary to simply increase electric power consumption and costs when increasing the computing for supercomputers. With the improvement of computing architecture and IC process, the electric power and cost efficiency can be improved to achieve high performance with less electric power and fewer cost. If we compared current fastest supercomputer Frontier, with Titan, which was the world's fastest in ten years ago, the performance of has increased up to 95.7 times, nearly a hundred times. At the same time, the consumed power is only increased by 2.6 times, while the cost is only 5.2 times. For details, see Fig.3 for the comparison of overall performance, cost, and electric power of supercomputers, and Fig.4 for comparison of computing performance per unit cost and unit electric power consumption. Besides, using of GPUs or AI Accelerators in supercomputer architecture will greatly improve AI training and Inference performance.

IV. SUPERCOMPUTER STATUS AND PLANS

Many countries are planning to build exa-scale supercomputers to meet huge computing requirements. The United States has the exa-scale Computing Project [6], which plans to build three exa-scale supercomputers before 2023. Besides, they released an RFI (Request for Information) [7] and the goal is to build 10-20 (FP64) exa-flops in 2025+ time frame, and 100+ (FP64) exa-flops computing power in 2030+ time frame. China appears to already have two exa-scale supercomputers and will build more in the future [8]. In the EU, EuroHPC JU (European High Performance Computing Joint Organization) will build an exa-scale supercomputer in Germany and three pre-exascale supercomputers in Finland, Spain and Italy [9]. Japan already has developed Fugaku supercomputer by themselves with a computing power up to 0.442 exaflops, which was fastest supercomputer in 2020. In addition, South Korea has the plan named "National Ultra High Performance Computing Innovation Strategy", which is expected to build an exa-scale supercomputer by 2030 [11]

V. ISSUES IN FOUNDATION MODEL

On the other hand, The pre-training method has become the mainstream to train large-scale neural network, In Pre-Train phase, the large-scale neural network first trains by huge amount of corpus to generate the AI Foundation Model, and then conducts downstream task training through by Fine Tune. The corpus is very important. If it contains biased or poisoned information, this flaw will affect all downstream AI tasks [12]. Therefore, how to avoid bias and poisoned information in AI foundation models has become a new information security issue. If we rely on an external large-scale AI neural network model, it is difficult to ensure the correctness of the corpus and the security of the AI model to meet our needs. It also reminds us of the importance of building our own capabilities and computing power to train large AI neural network models ourselves.

VI. SUMMARY

The parameter scale of large-scale AI neural network model has grown rapidly to pre-trillion-scale. There will be two directions for future trends: First, to enlarge NN parameters for higher performance or emergence of new features; and second, to reduce NN parameters to maintain same or acceptable performance to save computing power.

AI training computing pushes up computing needs to exa-scale, so many countries have exa-scale supercomputing projects. besides, AI inference for large-scale AI models also consumes a lot of computing power. Therefore, computational efficiency will be a key factor in the commercialization and popularization of AI applications.

AI development will be driven by the progress of computing power (supercomputer), algorithm (neural network model and parameter scale), and application (foundation model and downstream fine tuning). Large-scale AI neural network models demonstrate their potential to broadly impact human work and lifestyle. It is critical to have sufficient computing power and R&D capabilities for training and inference for them.

Fig.2 Training Compute (FLOPS) of milestone AI Neural Network Model over Time

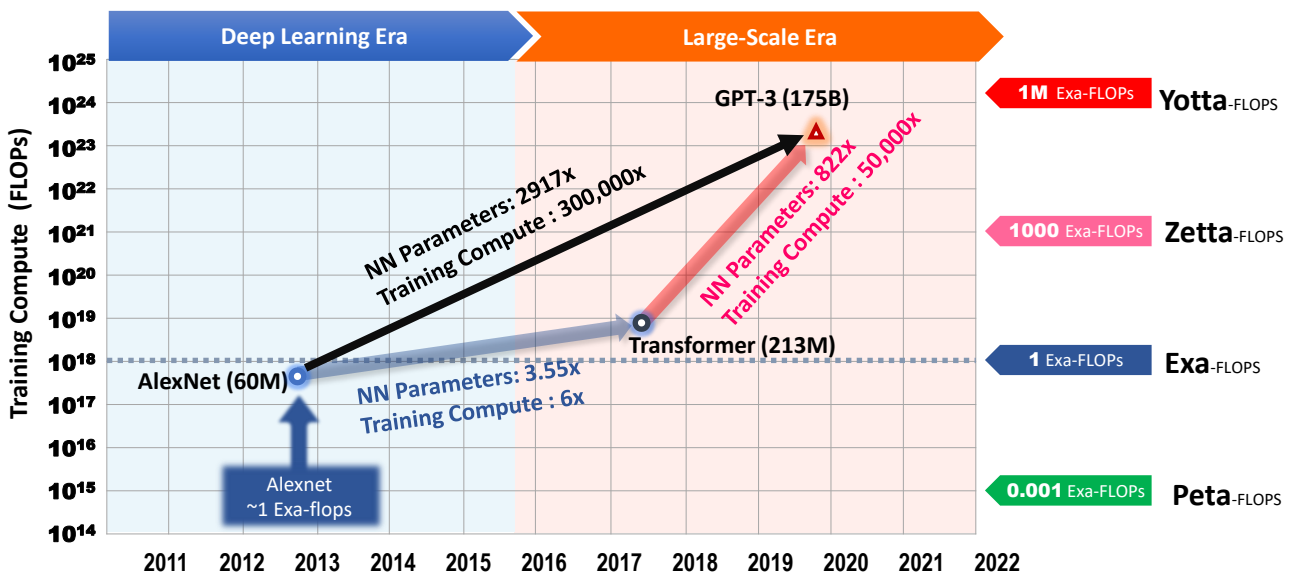


Fig 3. Comparison of overall performance, cost, and electric power of supercomputers (compared with Titan, the world's fastest supercomputer in 2012)

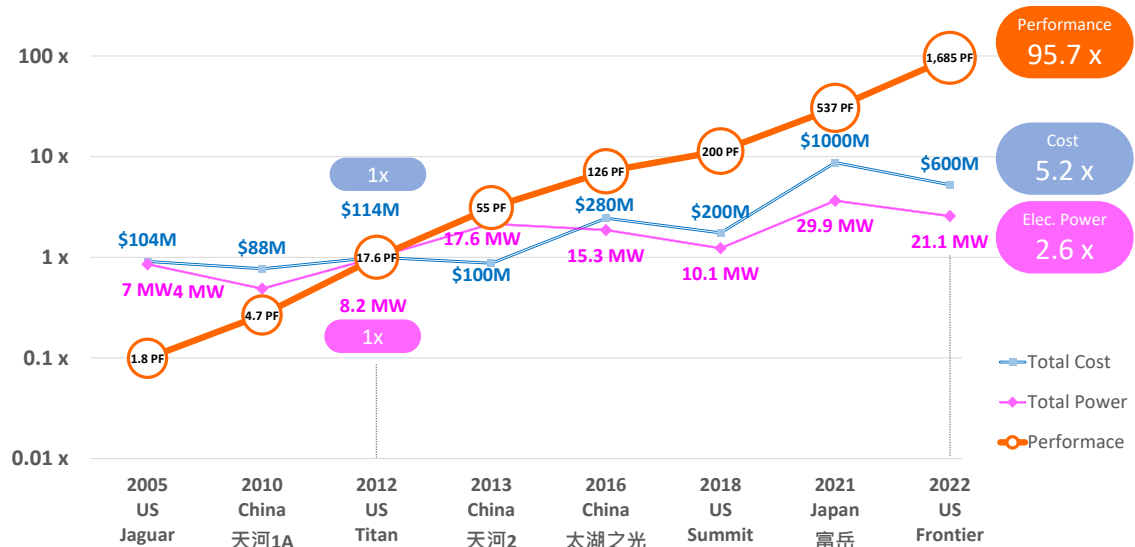
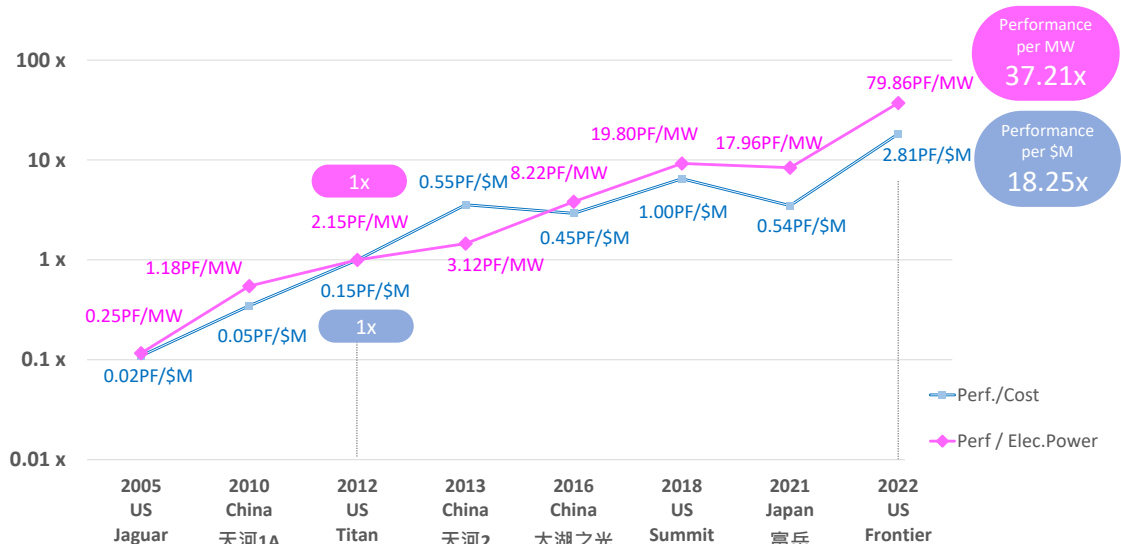


Fig.4. Comparison of computing performance per unit cost and unit electric power consumption (compared with Titan, the world's fastest supercomputer in 2012)



* PF = Peteflop/s , by benchmark Linpack (FP64) R-peak ; MW = MegaWatt

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